

SEQPOWERS

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Problem

Determine the set of all real numbers r for which there is an infinite sequence of positive integers $a_1, a_2, \dots$ with the following three properties:

- (1) No number occurs more than once as a sequence member.
- (2) The sum of two different sequence members is never a power of two.
- (3) $a_n < rn$ applies to all positive integers n .

Solution. All $r \geq 2$ work.

Say for r we have the sequence $b_1, b_2, \dots$. Then for $r_0 \geq r$, since $a_n < rn \leq r_0 n$, $b_1, b_2, \dots$ will work for r_0 as well.

So it is enough to show that $r = 2$ works.

Let $a_1 = 1$ and $a_2 = 2$ and $a_{2^{n-1}+1} = 2^n$ for all n . We will define the terms inductively and recursively.

Suppose we know the terms $a_1, a_2, \dots, a_{2^{n-1}}$, we will define the terms $a_{2^{n-1}+1}, \dots, a_{2^n+1}$. Consider the pairs of the form $(t, 2^{n+1} - t)$ for $t \in \{1, 2, \dots, 2^n\}$. We have 2^n pairs and we must have exactly one number from each pair.

Let $b_1, b_2, \dots, b_{2^{n-1}-1}$ be the numbers in increasing order that have not occurred in $a_1, a_2, \dots, a_{2^{n-1}}$ from the set $\{1, 2, \dots, 2^n - 1\}$. Then let $a_{2^{n-1}+k+1} = 2^{n+1} - b_{2^{n-1}-k}$. Observe that $2^{n+1} - b_{2^{n-1}-k} < 2^{n+1} < 2(2^{n-1} + k + 1)$ so it is valid.

Inductively, any number from $a_1, \dots, a_{2^{n+1}}$ does not sum to $1, 2, 4, \dots, 2^{n+1}$. Since every number is less than 2^n , they don't sum to anything $\geq 2^{n+2}$ as well. As a result, our construction works.

Now let $r = 2 - \varepsilon$ where $\varepsilon > 0$ is fixed. Say r works with the sequence $a_1, a_2, \dots$.

Let k be the smallest number such that k does not appear in the sequence $a_1, a_2, \dots$.

Consider the sequence $a_1, a_2, \dots, a_{2^{n-1}}$ and the pairs of the form $(t, 2^{n+1} - t)$ for $t \in \{1, 2, \dots, 2^n\}$. Observe that $a_{2^{n-1}} < 2^n$, so exactly one term of the sequence appears in the sequence. Since k does not appear, so $2^n - k$ appears in the sequence.

Since we have $a_m < rm < 2m$, so $2^n - k \in \left\{a_{2^{n-1}-\lfloor \frac{k}{2} \rfloor}, \dots, a_{2^{n-1}}\right\}$. Say $2^n - k = a_{2^{n-1}-j}$ where $j \in \{1, \dots, \lfloor \frac{k}{2} \rfloor\}$. So, $k = 2^n - a_{2^{n-1}-j} \geq 2^n - (2 - \varepsilon)(2^{n-1} - j) =$

$\varepsilon \cdots 2^{n-1} - (2 - \varepsilon)j = f(n)$. Since $\lim_{n \rightarrow \infty} f(n) \rightarrow \infty$, we get that k is unbounded. Since k is a fixed real, it is a contradiction! So $r < 2$ will not work!